

Francisco Mena

SCIENTIST · COMPUTER SCIENCE

Potsdam, Germany

✉ famenatoro@gmail.com | 📧 Francisco-Mena-3 | 📞 Francisco Mena | 🌐 fmenat | 📱 fmenat | 0000-0002-5004-6571 | 🐦 fmenat14

Research Lines & Interests

Deep Learning	Artificial Neural Networks, Multi-view Learning, Data Fusion, Multi-sensor Modeling, (Variational) Autoencoders
Applications	Earth Observation, Vegetation Applications, Crowdsourcing, Neural Information Retrieval, Astroinformatics
Unsupervised Learning	Dimensionality Reduction, Representation Learning, Latent Variable Modeling

Education

PhD in Computer Science

UNIVERSITY OF KAISERSLAUTERN-LANDAU (RPTU)

- Thesis title: *Data Fusion in Multi-view Learning for Earth Observation Applications with Missing Views*.
- *Magna Cum Laude*

Kaiserslautern, Germany

Jan. 2022 - May 2026

Magíster en Ciencias de la Ingeniería Informática

FEDERICO SANTA MARÍA TECHNICAL UNIVERSITY (UTFSM)

- Equivalent to *Master of Science in Computer Engineering*.
- Thesis title: *Mixture Models for Learning in Crowdsourcing Scenarios*.
- Grade Point Average: 94%.

Valparaíso, Chile

Mar. 2018 - Sep. 2020

Ingeniería Civil en Informática

FEDERICO SANTA MARÍA TECHNICAL UNIVERSITY (UTFSM)

- Equivalent to *Computer Engineering*.
- Grade Point Average: 80%.
- Top 10% on Class Rank. – Rank #4 of 66 students.

Santiago, Chile

Mar. 2013 - Sep. 2020

Licenciado en Ciencias de la Ingeniería Informática

FEDERICO SANTA MARÍA TECHNICAL UNIVERSITY (UTFSM)

- Equivalent to *Bachelor of Science in Computer Engineering*.
- Records linked to "*Ingeniería Civil en Informática*".

Santiago, Chile

Mar. 2013 - Nov. 2017

Experience

GFZ Helmholtz Centre for Geosciences

SCIENTIST

- Work on forest disturbance monitoring with remote sensing data, including resource and efficiency tracking.
- **Technologies:** Python, PyTorch, Slurm, GitLab, QGIS, and NextCloud.

Potsdam, Germany

Oct. 2025 - Now

German Research Centre for Artificial Intelligence (DFKI)

STUDENT RESEARCH ASSISTANT

- Part-time work on crop yield prediction and flood detection with remote sensing data.
- **Technologies:** Python, PyTorch, Slurm, GitLab, QGIS, Confluence, Jira, Teams, and OneDrive.

Kaiserslautern, Germany

Mar. 2022 - Sept. 2025

Inria

PHD VISITOR RESEARCHER

- Research exploration in topics of multi-modal co-learning, multi-task learning and mutual distillation.
- **Technologies:** Python, PyTorch, Slurm, Slack, and Overleaf.

Montpellier, France

Nov. 2024 - Jan. 2025

Federico Santa María Technical University (UTFSM)

RESEARCH ASSISTANT AT CHILEAN VIRTUAL OBSERVATORY (CHIVO)

- Work on the astronomical data reduction of ALMA and ESO observatories, and the creation of astronomical datasets.
- **Technologies:** Python, Jupyter Notebook, FITS, and Slurm.

Santiago, Chile

Jul. 2017 - May 2018

Farmacia Las Rosas S.A.

FRONT-END & BACK-END DEVELOPER

- Automation of refill stocks and database management.
- **Technologies:** Python, QT, and Excel.

Santiago, Chile

Jan. 2017 - Mar. 2017

University of Kaiserslautern-Landau (RPTU)

ACADEMIC

- (Lecturer) *Machine Learning for Earth Observation Applications* topic inside the Applications of Machine Learning and Data Science course.

Kaiserslautern, Germany

Oct. 2024 - Apr. 2025

Federico Santa María Technical University (UTFSM)

ACADEMIC

- (Lecturer) Computational Statistics, 3 times, since 2020.
- (Lecturer) Artificial Neural Networks, 1 time, in 2020.
- (Teacher Assistant) Computational Statistics, 2 times, since 2019.
- (Teacher Assistant) Artificial Neural Networks, 3 times, since 2018.
- (Teacher Assistant) Machine Learning, 3 times, since 2017.
- (Teacher Assistant) Fundamentals of Operations Research, 3 times, since 2017.
- (Laboratory Assistant) Mathematics, 1 time, in 2014.

Santiago, Chile

2014 - 2021

Honors & Awards

2022	PhD Scholarship, RPTU in Kaiserslautern	2022-2025
2019	Incentive Program for Scientific Initiation (PIIC), Federico Santa María Technical University	2019-2020
2018	Master program scholarship, Federico Santa María Technical University	2018-2020
2013	Honor Roll, Institutional excellence, Federico Santa María Technical University	2013

Research funding

2020	Investigator, DGIP PL_M_17_6, Federico Santa María Technical University (UTFSM)	Chile
2019	Research Assistant, BASAL FB-0008, Advanced center for Electrical & Electronic Engineering (AC3E)	Chile
2017-2018	Research Assistant, FONDEF IT15I10041, Chilean Virtual Observatory (ChiVO)	Chile

Selected Publications

PEER-REVIEWED ARTICLES IN JOURNAL

Multi-modal co-learning for Earth observation: enhancing single-modality models via modality collaboration	Machine Learning
F MENA, D IENCO, C F. DANTAS, R INTERDONATO, A DENGEL * DOI 10.1007/s10994-025-06903-0	Nov. 2025
In the search for optimal multi-view learning models for crop classification with global remote sensing data	Int. J. Appl. Earth Obs. Geoinf.
F MENA, D ARENAS, A DENGEL * DOI 10.1016/j.jag.2025.104823	Sep. 2025
Missing data as augmentation in the Earth Observation domain: A multi-view learning approach	Neurocomputing
F MENA, D ARENAS, A DENGEL * DOI 10.1016/j.neucom.2025.130175	Jul. 2025
Adaptive fusion of multi-modal remote sensing data for optimal sub-field crop yield prediction	Remote Sensing of Environment
F MENA, D PATHAK, H NAJJAR, C SANCHEZ, P HELBER, B BISCHKE, P HABELITZ, M MIRANDA, ET AL. * DOI 10.1016/j.rse.2024.114547	Mar. 2025
Common practices and taxonomy in deep multiview fusion for remote sensing applications	IEEE JSTARS
F MENA, D ARENAS, M NUSKE, A DENGEL * DOI 10.1109/JSTARS.2024.3361556	Feb. 2024
On the quality of deep representations for Kepler light curves using variational auto-encoders	MDPI Signals
F MENA, P OLIVARES, M BUGUEÑO, G MOLINA, M ARAYA * DOI 10.3390/signals2040042	Oct. 2021
Collective annotation patterns in learning from crowds	Intelligent Data Analysis
F MENA, R ÑANCULEF, C VALLE * DOI 10.3233/IDA-200009	Dec. 2020

PEER-REVIEWED CONFERENCES

Co-Learning for missing arbitrary modalities in multi-modal classification	Discovery Science, Springer
F MENA, D IENCO, R INTERDONATO, C F. DANTAS, S BESNARD * Upcoming	Oct. 2026
YieldSAT: A multimodal benchmark dataset for high-resolution crop yield prediction	CVPR, IEEE
M MIRANDA, D PATHAK, P HELBER, B BISCHKE, H NAJJAR, F MENA, C SANCHEZ, A PAI, D ARENAS, ET AL. * Upcoming	Jun. 2026
An analysis of temporal dropout in Earth observation time series for regression tasks	IDA, Springer
M MIRANDA, F MENA, M CHARFUELAN, A DENGEL * DOI 10.1007/978-3-031-91398-3_29	May 2025
Impact assessment of missing data in model predictions for Earth observation applications	IGARSS, IEEE
F MENA, D ARENAS, M CHARFUELAN, M NUSKE, A DENGEL * DOI 10.1109/IGARSS53475.2024.10640375	Oct. 2024
A binary variational autoencoder for hashing	CIARP, Springer
F MENA, R ÑANCULEF * DOI 10.1007/978-3-030-33904-3_12	Oct. 2019

Skills

Computer	Python, Keras, PyTorch, TensorFlow, Slurm, Visual Studio Code, Jupyter Notebook, LaTeX, C++, C, R, Sony Vegas.
Mentoring / Assessment	Five Master's theses, five student projects, reviewer of various journals
Communication	Lecturer role at University, session chair at conferences, presentations at conferences and workshops.
Coordination	Conference workshop co-organization, organizer of conference sessions, research exchange sessions.
Personal	Teamwork, planning and organization, responsibility, collaboration.
Languages	Spanish (Native), English (IELTS 7.0), Portuguese (Basic).

References

Martin Herold	martin.herold@gfz.de , Remote Sensing and Geoinformatics, GFZ Helmholtz Centre for Geosciences.
Andreas Dengel	andreas.dengel@dfki.de , Department of Computer Science, University of Kaiserslautern-Landau (RPTU), and DFKI GmbH.
Dino Ienco	dino.ienco@inrae.fr , INRAE, Inria, and University of Montpellier.